

TAXONOMIC COMPLETENESS AND KNOWLEDGE GAPS IN QUEENSLAND DEEP SEA ECHINODERMS ACROSS A DEPTH GRADIENT

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Background

Deep-sea biodiversity remains far less studied and taxonomically resolved than shallow-water fauna.

Why it matters

Deep-water collections are not only under-sampled but under-resolved.

Using integrated museum and biodiversity records, we assessed taxonomic completeness and knowledge gaps in Queensland echinoderm collections across a depth gradient.

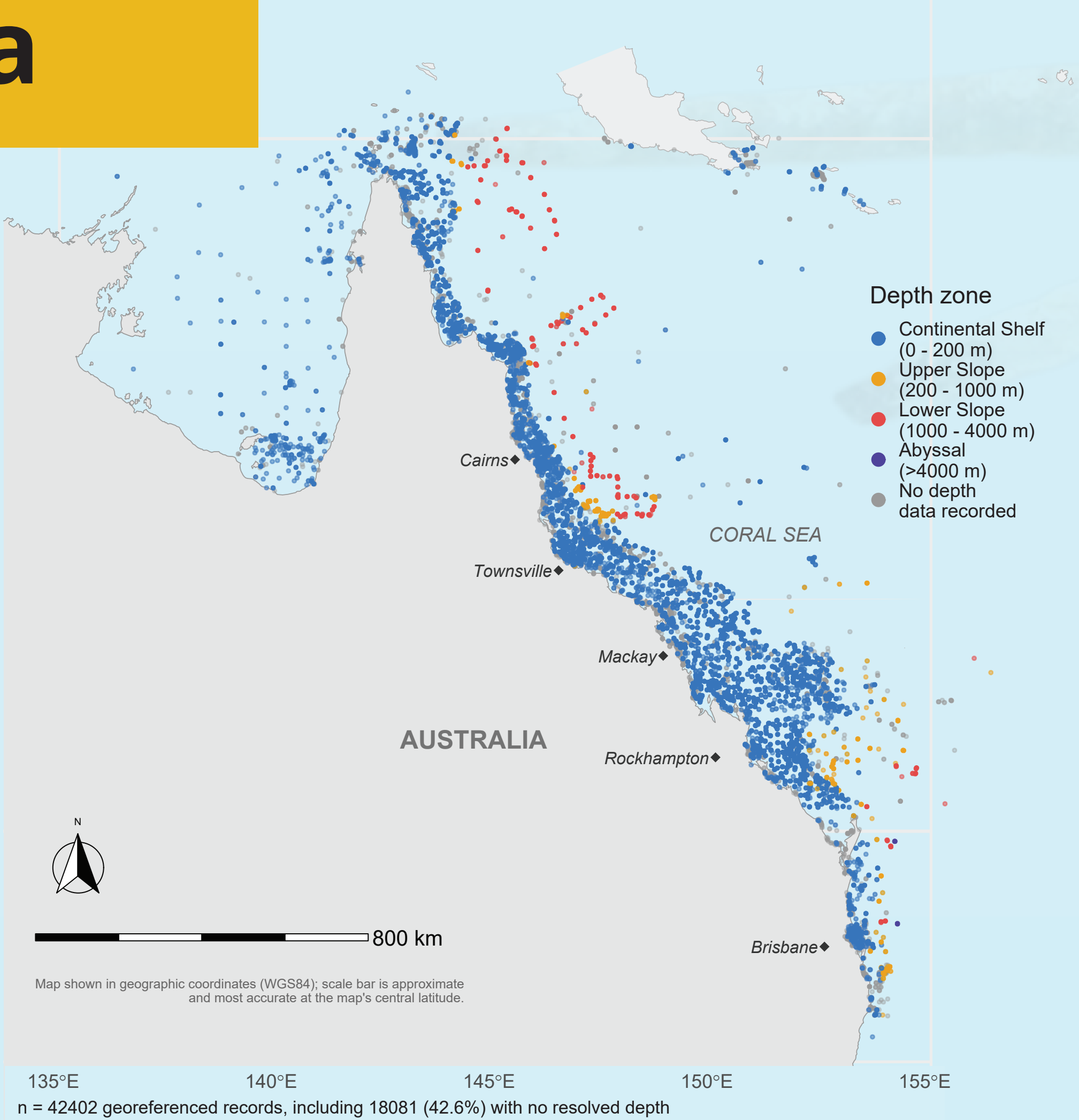
Improving species-level identification of existing museum collections may reveal substantial biodiversity knowledge without additional collecting effort.

Study Area

Echinoderm occurrence records across the study region (all records)

Year Range: 1861-2026
Depth range: 0 - 4,280 m
species: 930

Figure 1. Echinoderm occurrence records across the study region by depth zone. Only georeferenced records passing coordinate quality control are shown (n=42,402 of 43,470 validated records).



Dataset & Workflow

Atlas of Living Australia
ala.org.au

iDigBio
Integrated Digitized Biocollections

O BIS
OCEAN BIOGEOGRAPHIC INFORMATION SYSTEM

AUSTRALIAN MUSEUM

CSIRO

QM TROPICS

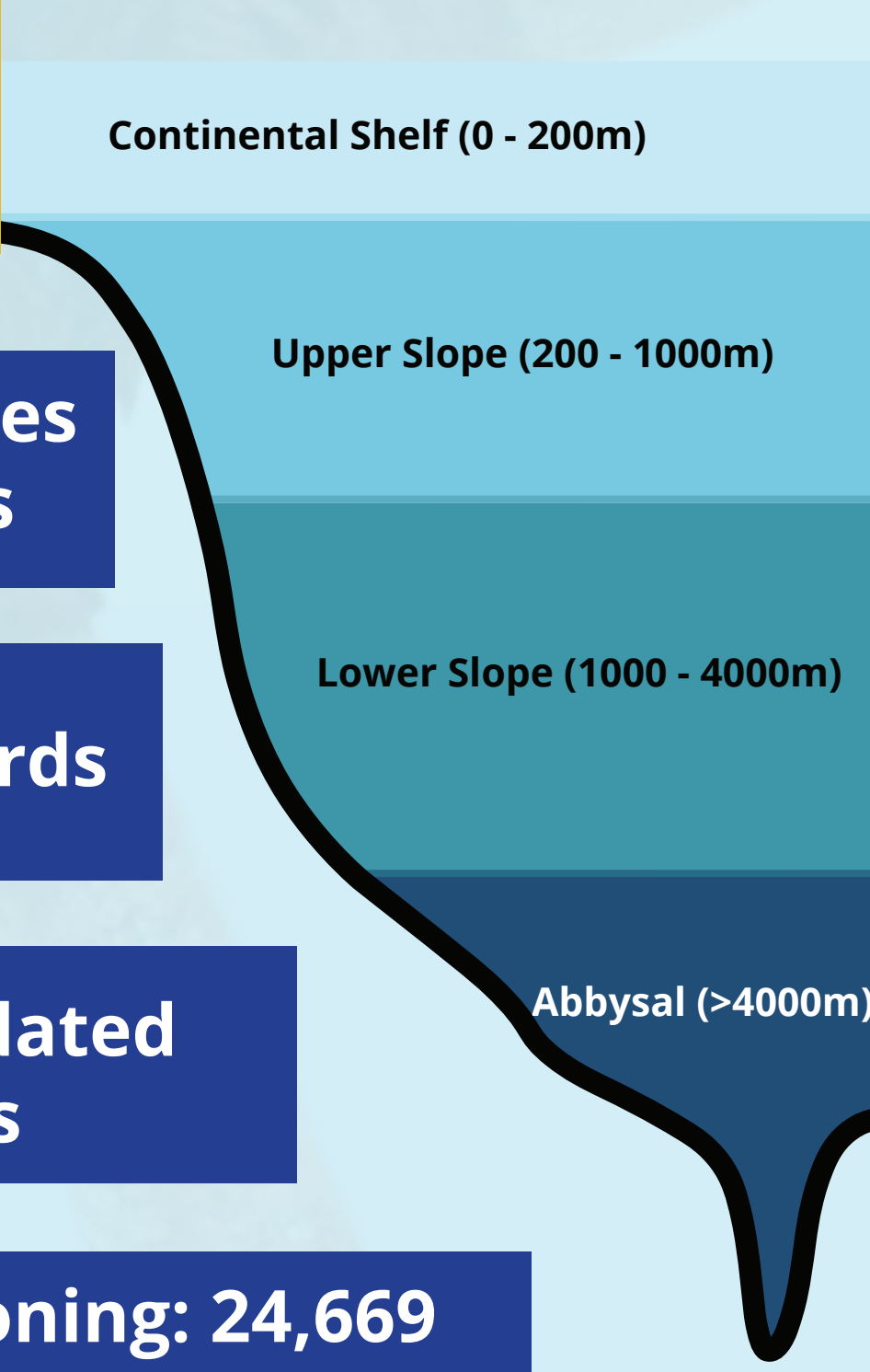
DATA PUBLISHER - GBIF - DATA PUBLISHER - GBIF

13 Biodiversity Databases & Museum Collections

459,956 raw records

43,470 validated records

Depth zoning: 24,669 depth-assigned records



Direct collection exports provided by M. Mitchell, S. Borghi (QMT) and C. Rowe (AM).

Key Findings

How complete is our knowledge of Queensland's echinoderm fauna across the depth gradient?

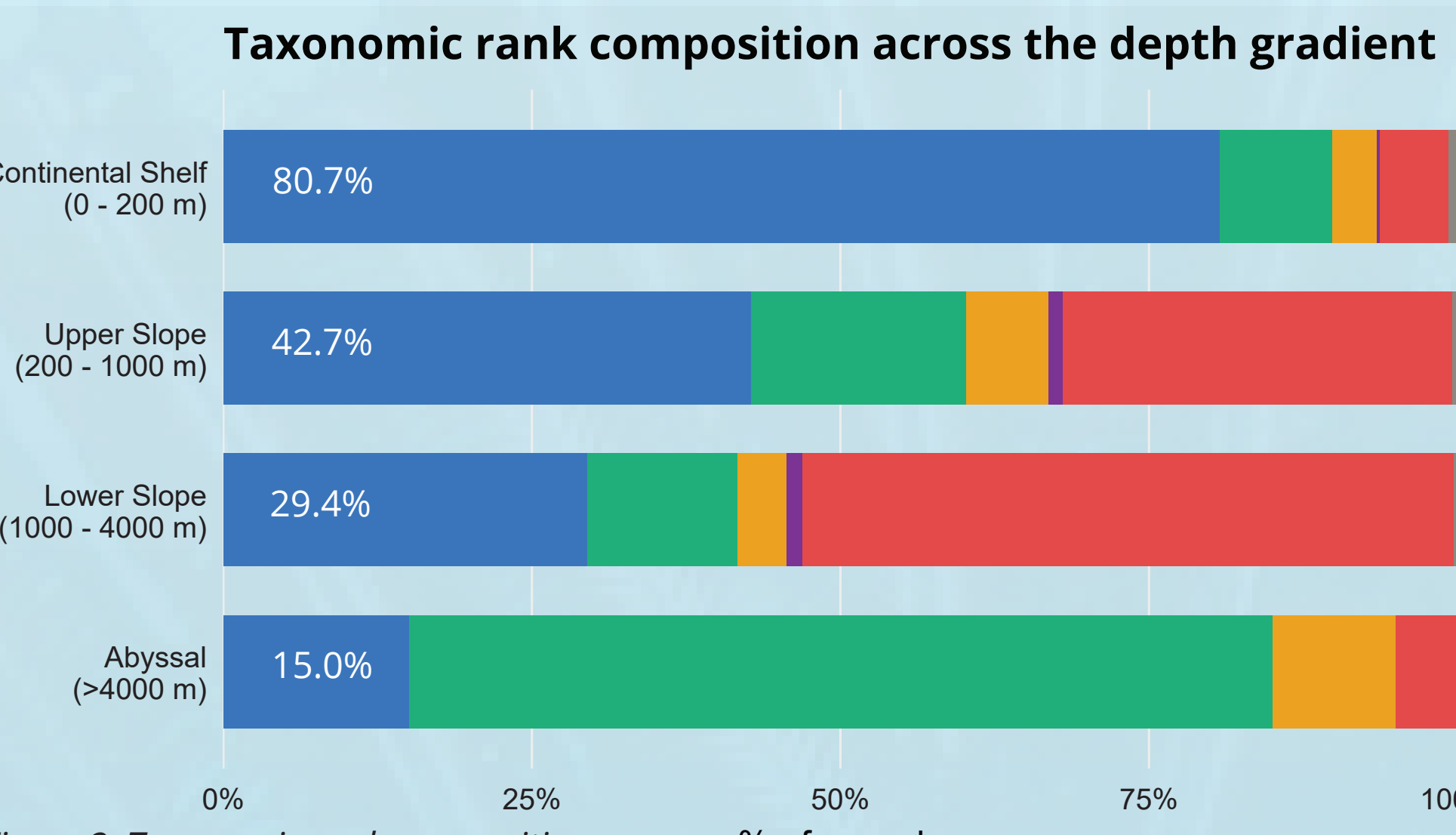


Figure 2. Taxonomic rank composition across the depth gradient. Rank = highest taxonomic level to which each record is resolved.

How well do collections represent regional diversity?

Collection captured 701 of 873 regionally documented echinoderm species, giving an overall TRI of 80.3%.

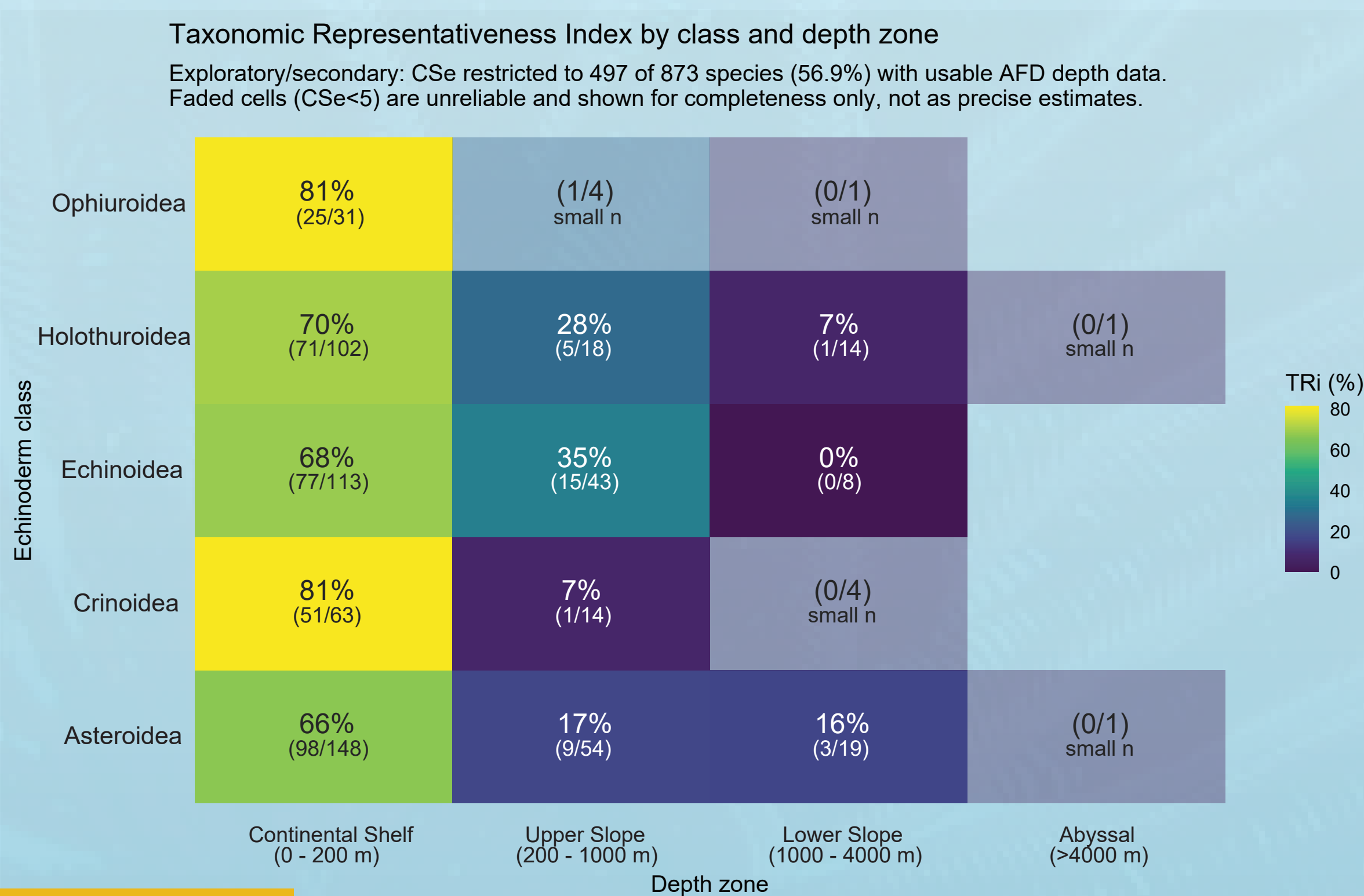


Figure 3. Taxonomic Representativeness Index (TRI) by echinoderm class and depth zone. TRI = regional species total (Australian Faunal Directory, AFD) and CSI = species recorded in this collection. Faded cells: CSe < 5.

Conclusion

- Species-level identification declined from 80.7% (shelf) to 15% (abyssal).
- Echinoidea showed the highest identification rates; Holothuroidea the lowest.
- Slope and abyssal collections are the highest priority for future taxonomic work.

Where are the major taxonomic knowledge gaps?

Most Taxonomically Resolved Echinoidea: Sea Urchin

Prionocidaris australis

Least Taxonomically Resolved Holothuroidea: Sea Cucumber

Peniagone sp

Himerometra robustipinna

Comparable identification between them

Crinoidea: Feather Star

Ophiuroidea: Brittle Star

Asteroidea: Sea Star

Lower Slope and Abyssal zones contain the greatest taxonomic knowledge gaps.

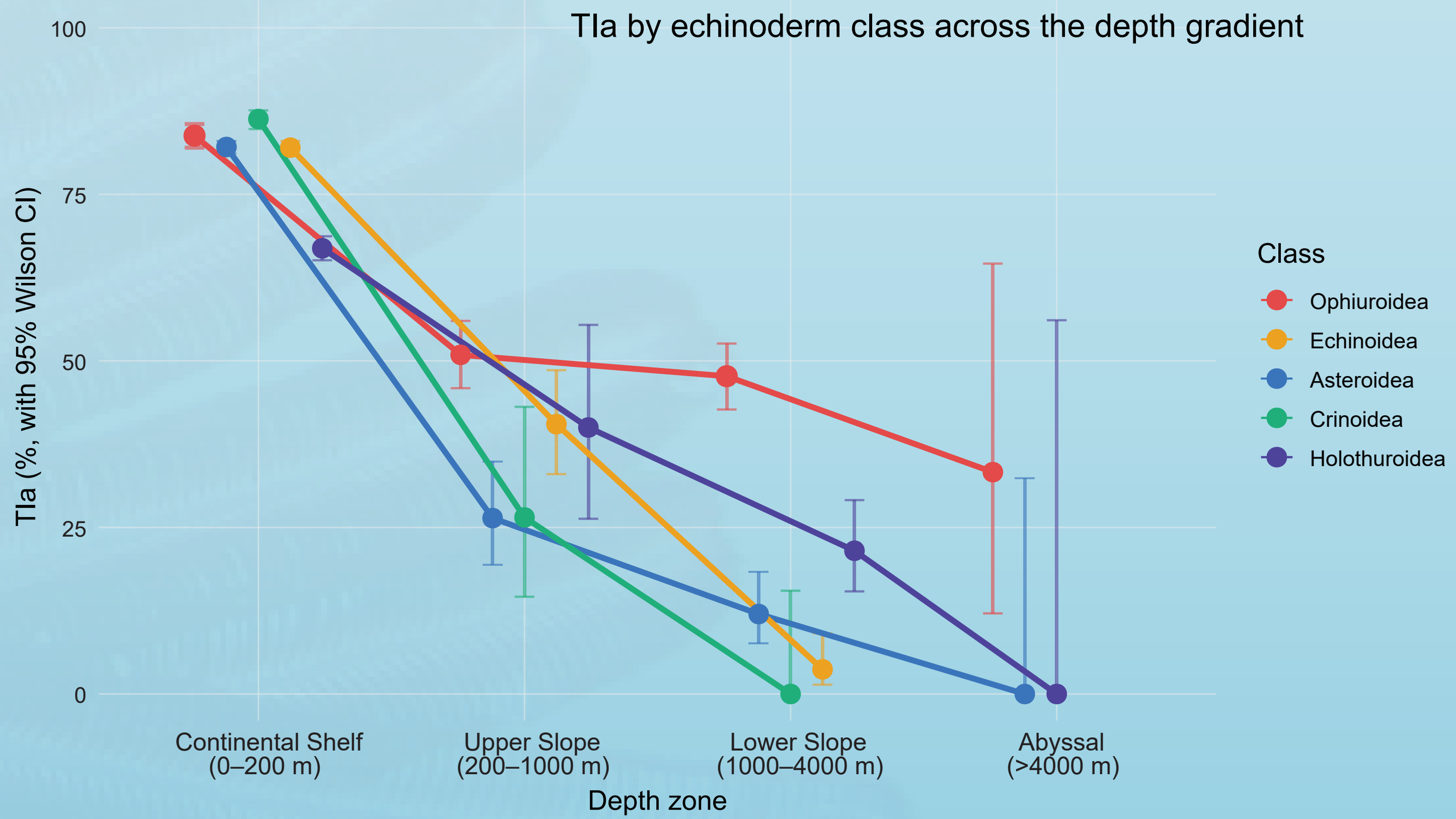


Figure 4. Taxonomic Identification Index (TII) by echinoderm class across the depth gradient, with 95% Wilson confidence intervals.

Reference

